

Econ 440: Potential Outcome Framework

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1 Introduction

In these notes, we define what a causal effect is, and what our causal parameters of interest will be (i.e., what we want to estimate). We also discuss the main challenges to estimate those parameters. We do this using the *Potential Outcome Framework*.

- The Potential Outcome Framework is also called the Rubin Causal Model (RCM)
- It has its roots in the analysis of experiments by Neyman and Fisher in the 1920's.
 - It was later formalized and extended to non-experimental settings by Donald Rubin in the 1970's.
- It is a widely used approach in economics and many other fields to study causality.
- It is helpful to formally define causal effects and study their estimation.

There are Two Key Elements of this Framework: $\begin{cases} 1. \text{ Potential Outcomes} \\ 2. \text{ Assignment Mechanism} \end{cases}$.

We will go over them in these notes, and also define what a “Causal Effect” is.

2 Setting

Goal: Study the causal effect of a “treatment” (T) on an outcome (Y).

- In this framework, a “treatment” is an *action or intervention* applied to a *unit* (e.g. an individual or firm).

Examples of Treatments:

- Attend college or not

- Take a drug or not
 - Enroll in a training program or not
 - Receive welfare
 - Smoke or Don't Smoke
- Let “ i ” denote a given individual/unit from our population of interest (e.g. workers in the U.S.)
 - For now, we focus on a *binary* treatment:

Let $T_i = \{0, 1\}$ indicate whether “ i ” got the treatment ($T_i = 1$) or not ($T_i = 0$).

For example, let

$$T_i = \begin{cases} 1 & \text{if individual } i \text{ attended college} \\ 0 & \text{if not} \end{cases}.$$

3 Potential Outcomes

Potential outcomes are the first key element of the potential outcome framework.

Given a unit i and set of actions $T = 1$ and $T = 0$, we associate each unit-action pair with a potential outcome.

Definition. Given a unit i and set of actions $T = 1$ and $T = 0$, define the *potential outcomes* of unit i as:

- $Y_i(1)$: The outcome individual i would obtain if i were to get treatment.
- $Y_i(0)$: The outcome individual i would obtain if i were to not get treatment.

Example:

- $Y_i(1)$: Earnings of i if he/she had attended college.
- $Y_i(0)$: Earnings of i if he/she had not attended.

Notes:

- Also written as: Y_i^1 , $Y_{1,i}$, $Y(T_i = 1)$ or Y_i^0 , $Y_{0,i}$, $Y(T_i = 0)$.

- They are “potential” in the sense that only one of them will be realized/observed, depending on the action taken:
 - If $T_i = 1 \implies Y_i(1)$ is observed
 - If $T_i = 0 \implies Y_i(0)$ is observed
- Hence, the outcome we actually observe for individual i , or i ’s *observed outcome* denoted by Y_i^{obs} , is given by:

$$Y_i^{obs} = \begin{cases} Y_i(1) & \text{if } T_i = 1 \\ Y_i(0) & \text{if } T_i = 0 \end{cases}.$$

or,

$$Y_i^{obs} = T_i Y_i(1) + (1 - T_i) Y_i(0).$$

Usually, we write just “ Y_i ” instead of Y_i^{obs} .

- In general, the data we observe is: $\{T_i, Y_i\}$.
- In this course, and causal inference in general, it is *very important* to always keep in mind the distinction between *potential* and *observed* outcomes.

4 Causal Effects

Definition. A *Causal Effect* is a comparison of potential outcomes for the same unit, or same set of units.

Definition. The *Individual Causal Effect* of T on Y for unit i is given by:

$$ICE_i = Y_i(1) - Y_i(0).$$

Example: The ICE_i of attending college on earnings for individual i equals i ’s earnings had she/he attended college minus i ’s earnings had she/he not attended college.

Notes:

- The definition does not depend on which outcome we *observe*; that is, it is not a function of Y_i^{obs} .
- The definition does not depend on which *estimation method* we use (e.g OLS, IVs, etc.)

4.1 Key Challenge to Estimate Causal Effects: The FPCI

Why is it difficult to estimate causal effects?

Fundamental Problem of Causal Inference (FPCI) (Holland, 1986): We can *never* observe both potential outcomes for the *same* individual at the *same* time.

Notes:

- We observe either $Y_i(1)$ if $T_i = 1$ or $Y_i(0)$ if $T_i = 0$, never both.

Remember:
$$Y_i^{obs} = \begin{cases} Y_i(1) & \text{if } T_i = 1 \\ Y_i(0) & \text{if } T_i = 0 \end{cases}.$$

- The *unobserved* potential outcome for unit i is usually called the “*counterfactual outcome*”. It equals:

$$\begin{aligned} &Y_i(0) \text{ if } T_i = 1 \\ &Y_i(1) \text{ if } T_i = 0 \end{aligned}$$

- The FPCI implies that estimation of causal effects involves making inferences about what those counterfactual or hypothetical outcomes would be $\implies$ It is hard to estimate causal effects!
- We can think of the problem of estimating causal effects as a *missing data* problem:

$$Y_i(1) = \begin{cases} \text{missing} & \text{if } T_i = 0 \\ Y_i^{obs} & \text{if } T_i = 1 \end{cases}$$

$$Y_i(0) = \begin{cases} Y_i^{obs} & \text{if } T_i = 0 \\ \text{missing} & \text{if } T_i = 1 \end{cases}$$

$\implies$ Therefore, it is extremely hard to estimate the Individual Causal Effect $ICE_i = Y_i(1) - Y_i(0)$.

- Due to the FPCI and the difficulty to estimate Individual Causal Effects, we usually work with many units and focus on other causal effects (or quantities), such as *average*, median, or quantile effects. These causal effects are still very hard to estimate, but at least not as much as individual causal effects.

4.2 Causal Parameters of Interest: ATE and ATT

In what follows, we assume we have a random sample $i = 1, \dots, n$ from our population of interest (e.g. sample of US workers).

Question: What do we want to estimate?

- We typically focus on *Average Causal Effects*. They are easier to estimate than individual causal effects.
- The two most common parameters of interest are:
 1. The Population Average Treatment Effect (*ATE*)
 2. The Average Treatment Effect on the Treated (*ATT*)

4.2.1 Average Treatment Effect: ATE

Definition. The *Population Average Treatment Effect (ATE)* is defined as:

$$ATE = E[Y_i(1) - Y_i(0)] = E[Y_i(1)] - E[Y_i(0)] \quad (1)$$

Notes:

- It gives the average of the individual causal effects (ICE_i) *in the population*, i.e., $ATE = E[Y_i(1) - Y_i(0)]$.
- It also gives the difference of the average potential outcome had all individuals in the population of interest received the treatment (i.e., $E[Y_i(1)]$) minus the average potential outcome had all individuals not received the treatment (i.e., $E[Y_i(0)]$).
- It answers: what is the average causal effect of T on Y in the population?
 - In our leading example, it answers: What is the average effect of attending college on earnings for all US workers (our population of interest)?
- Remember: The *ATE* is not a random variable; it is a fixed (but unknown) quantity.

4.2.2 Average Treatment Effect on the Treated: ATT

Definition. The *Average Treatment Effect on the Treated (ATT)* is defined as:

$$ATT = E[Y_i(1) - Y_i(0)|T_i = 1] \quad (2)$$

Notes:

- It gives the average (individual) causal effect *for those units or individuals who received treatment*.
- We can also write it as:

$$ATT = E[Y_i(1) - Y_i(0)|T_i = 1] = E[Y_i(1)|T_i = 1] - E[Y_i(0)|T_i = 1]$$

Given that $Y_i = Y_i(1)$ for units with $T_i = 1$, we have that:

$$ATT = E[Y_i|T_i = 1] - E[Y_i(0)|T_i = 1]$$

- It gives the difference in the average observed outcome of those who get treatment minus the average of the potential outcome they would have obtained had they not received treatment.
- It answers: what is the average causal effect of T on Y for those who received treatment?
 - In our leading example, it answers: What is the average effect of attending college on earnings *for those who attended college*.
 - Note that those who attended college are a sub-population, and they may be different from those who did not attend college, and also from the entire population.
 - Remember: Those who attend college may be a self-selected group; on average, they may have higher socioeconomic status, higher ability, motivation, etc.
 - In other words, in this example, those who attend college are not “similar” to (or representative of) the overall population.

4.2.3 ATE vs. ATT

Both parameters are important and widely used in the literature and in practice. Which one to use depends on our question of interest. Examples:

- Mean effect of receiving food stamps on the health of those who receive them? *ATT*

- Mean effect of receiving food stamps on economically disadvantaged people—our population of interest (some of whom may or may not receive food stamps)? *ATE*
- Mean effect of attending college on earnings for all US workers? *ATE*
- Mean effect of attending college on earnings for those who attended college? *ATT*
- If we want to do a cost/benefit analysis of a current program: Use *ATT*, as we want the benefit for those who are participating.
- Mean effect of a training program on its “target population” (e.g., economically disadvantaged youth): *ATE*
- Mean effect of a training program on its actual participants (i.e., on those who enrolled in it): *ATT*

In what follows, these are the causal effects we want to estimate (*ATE* and *ATT*).

4.2.4 Final Note on ATE and ATT

The set of units who receive treatment (i.e. those with $T_i = 1$) is called the *Treatment Group*, and the set of units who does not receive treatment (i.e. those with $T_i = 0$) is called the *Control Group*.

5 Challenges to Estimate the ATE and ATT

5.1 ATE

From equation (1), we can write the *ATE* as:

$$ATE = E[Y_i(1)] - E[Y_i(0)]$$

Applying the Law of Iterated Expectations (LIE) to each of the two terms above, we obtain:

$$ATE = E_T[E[Y_i(1)|T_i]] - E_T[E[Y_i(0)|T_i]]$$

Then, we can expand the outer expectation (the one for T) in each of those terms to get:

$$\begin{aligned} ATE &= Pr(T_i = 0)E[Y_i(1)|T_i = 0] + Pr(T_i = 1)E[Y_i(1)|T_i = 1] \\ &\quad - \{Pr(T_i = 0)E[Y_i(0)|T_i = 0] + Pr(T_i = 1)E[Y_i(0)|T_i = 1]\} \end{aligned}$$

Finally, remember that $Y_i = Y_i(0)$ for units with $T_i = 0$, and $Y_i = Y_i(1)$ for units with $T_i = 1 \implies E[Y_i(0)|T_i = 0] = E[Y_i|T_i = 0]$ and $E[Y_i(1)|T_i = 1] = E[Y_i|T_i = 1]$. Therefore, we can write the ATE as:

$$ATE = Pr(T_i = 0)E[Y_i(1)|T_i = 0] + Pr(T_i = 1)E[Y_i|T_i = 1] - Pr(T_i = 0)E[Y_i|T_i = 0] - Pr(T_i = 1)E[Y_i(0)|T_i = 1] \quad (3)$$

Notes on equation(3):

- Given observed data $\{T_i, Y_i\}$ for $i = 1, \dots, n$, we can estimate (or “identify”): $Pr(T_i = 1)$, $Pr(T_i = 0)$, $E[Y_i|T_i = 1]$, and $E[Y_i|T_i = 0]$.
- *However*, to estimate the ATE , we are left with *two Missing Counterfactuals*:
 1. $E[Y_i(0)|T_i = 1]$: Mean potential outcome for *treated* group if they had not been treated.
 - Example: Average earnings under not attending college ($Y_i(0)$) for individuals who attended college ($T_i = 1$).
 2. $E[Y_i(1)|T_i = 0]$: Mean potential outcome for *control* group if they had been treated.
 - Example: Average earnings under attending college ($Y_i(1)$) for individuals who did not attend college ($T_i = 0$).
- Any observed data has no explicit information on those counterfactual outcomes due to the FPCI!
- Goal: Construct those counterfactuals to estimate the $ATE \implies$ Any approach/research design to estimate causal effect must answer:
 - (i) How are we constructing those counterfactuals?
 - (ii) What assumptions are we imposing to do that?
- Note: In general: $E[Y_i(1)|T_i = 0] \neq E[Y_i(1)|T_i = 1] = E[Y_i|T_i = 1]$, and $E[Y_i(0)|T_i = 1] \neq E[Y_i(0)|T_i = 0] = E[Y_i|T_i = 0]$.
 - *Why?*
Control and Treatment groups (i.e., groups with $T_i = 0$ and $T_i = 1$) are not “comparable” or similar. It is not a *ceteris paribus* comparison.

- Example: Those attending college have, on average, higher ability, socioeconomic status, etc.

5.2 ATT

For the ATT , we have a similar problem to the one we had for the ATE . We can write the ATT as:

$$ATT = E[Y_i(1) - Y_i(0)|T_i = 1]$$

Then, using the linearity property of (conditional) expectations:

$$ATT = E[Y_i(1)|T_i = 1] - E[Y_i(0)|T_i = 1]$$

Finally, using the fact that $Y_i = Y_i(1)$ for units with $T_i = 1$, we can write the ATT as:

$$ATT = E[Y_i|T_i = 1] - E[Y_i(0)|T_i = 1] \quad (4)$$

Notes on equation(4):

- The ATT has only *one* missing counterfactual, $E[Y_i(0)|T_i = 1] \implies$ It is in general (a little) easier to estimate than the ATE .
 - Intuition: In the context of our example, here we want the effect of “college” on earnings only for those who attended college, and we observe $Y_i(1)$ for all of them.
 - In this case, we do not “care” about the effect for the control group (those with $T_i = 0$), and hence, we do not care about finding $E[Y_i(1)|T_i = 0]$.

In this class, we will study approaches to estimate the ATE and the ATT . First, we will look at the Experimental Design (the “Gold Standard”).

6 Assignment Mechanism

The second key element of the Potential Outcome Framework is the Assignment Mechanism, which answers the question: how whether individual i gets $T_i = 0$ or $T_i = 1$ is determined? In other words, how is the treatment T assigned?

- The *Assignment Mechanism* formalizes how each unit gets either $T_i = 1$ or $T_i = 0$.
 - Examples: how is “college attendance” assigned? How is “food stamps” assigned? How is “Medicaid” assigned?
- The Assignment Mechanism plays a *key role* when estimating causal effects: What *research design* or tool we use depends on how T was assigned. Examples:
 - If T is assigned at random (e.g., by a lottery) $\implies$ use experimental design.
 - If T is assigned based on variables that we observe (e.g., age, SAT scores, income, etc.) $\implies$ use a “selection on observables” approach (e.g. OLS, matching).
 - If T is assigned based on variables that are unobserved to us (e.g., ability, motivation, etc.) $\implies$ use a “selection on unobservables” approach (e.g., IV).
- Economic Theory and Institutional Knowledge play a key role in understanding how T is assigned.

For our leading example with T = attending college: What determines who attends college ($T = 1$)? How is “attendance to college” assigned?

- Economics Theory: It is determined by cost/benefit analysis, opportunity costs, etc.
- Institutional Knowledge: How do scholarships work? What is the application process? etc.

7 Final Comments on Potential Outcome Framework

7.1 Covariates and Pre-Treatment Variables

- In applications, we usually have a set of covariates or “control variables”, denoted by X .
- In this framework, the variables X are (or must be) variables that are not affected by the treatment T . They are usually measured at baseline or before treatment is received. Due to this, they are also called *pre-treatment* variables. Examples:

- Ability, age, socioeconomic status, race, and gender before attending college.
- Education, age, experience, employment status before enrolling in a training program.
- In this approach, if a variable S is affected by T , then it cannot be a covariate or pre-treatment variable.
 - Examples: employment status, ability after attending college.
 - *Why is this the case?* If a variable S can be affected by T , it can then be an *intermediate outcome* or *mediator/mechanism/channel* $\implies$ Analysis gets a lot more complicated! (See, e.g., the papers by Flores and Flores-Lagunes (2009, 2010) and references therein for further details).

7.2 Stable Unit Treatment Value Assumption

To simplify our analysis, in this course we use the next commonly used assumption:

Assumption. The *Stable Unit Treatment Value Assumption (SUTVA)* (Rubin, 1974, 1978) states that:

1. The potential outcomes for any unit do not depend on other unit's treatment assignment (*"No interactions or spillover effects"*).
2. For each unit, there are no different versions of the treatment (*"Homogeneous treatment"*).

Notes:

- Without SUTVA, we would have *more* than two potential outcomes for each unit i .

Example:

With 2 units (i and j) and allowing for interaction or spillover effects among them, we would have the following potential outcomes for unit i :

1. $Y_i(1, T_j = 1)$
2. $Y_i(1, T_j = 0)$
3. $Y_i(0, T_j = 0)$
4. $Y_i(0, T_j = 1)$

For example, $Y_i(1, T_j = 0)$ would represent i 's earning if she had attended college and j did not attend college. With interactions effects, this potential outcome may differ from $Y_i(1, T_j = 1)$: i 's earning if she had attended college and j had also attended college.

Under SUTVA, we have:

1. $Y_i(1, T_j = 0) = Y_i(1, T_j = 1) = Y_i(1)$
2. $Y_i(0, T_j = 0) = Y_i(0, T_j = 1) = Y_i(0)$

Hence, we get the same potential outcomes as before: $Y_i(1)$ and $Y_i(0)$.

- It is important to keep in mind that this assumption may not hold in some applications. Examples:
 - Colleges may vary by quality, or “attending college” may differ by major $\implies$ There may be “different versions” of the treatment “attending college”.
 - Peer-effects in education (who your classmates are) may affect the outcomes.
 - A training program may have different components (e.g., different classes, or services like counseling, health services and job search assistance—not everybody may take the same components) $\implies$ There may be “different versions” of the treatment “enrolling in a training program”.
- Under SUTVA, we will have only two potential outcomes for each unit: $Y_i(1), Y_i(0)$.